

BINAY—Lighting Up The Future

Binay LED Aviation Obstruction Lights



BINAY LED-based MEDIUM INTENSITY
Aviation Obstruction Light Beacon

As per the International Civil Aviation Organisation (ICAO) requirements
for use on structures above 45m

Available in Low Intensity and Medium Intensity versions (as per
the International Civil Aviation Organization guidelines), BINAY's
patented LED Aviation Lights come with a 5-year warranty

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave, and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED
PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT



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chips is a small square-shaped silicon piece that vibrates when the chip is in motion. Below it a piezoelectric material is used, which when strained produces electric potential and is then used to produce electric current. This piezoelectric current is then passed through an electric device, which converts the electric current into a compatible form for the batteries. In other words, this device simply converts the vibrational energy into electrical energy by means of a piezoelectric material. For instance, when this chip assembly is associated with a rotating tire, it can produce about 200-microwatt power.

Solar bonsai tree to charge gadgets

Taking cue from the concept of photosynthesis, where trees use solar energy to prepare food for themselves, French designer Vivien Muller has designed an 'Electree' that can



Electree—a solar bonsai tree that can charge gadgets

feed gadgets. The Electree is a solar bonsai tree with 27 amorphous silicon solar panels sized 96x96 mm² that capture sunlight more efficiently.

The 40cm high Electree can be placed indoor near windows to get most of the sunlight. The panels harvest solar energy and store it in a 13,500mAh battery in the base of the bonsai tree. The battery can be fully charged within 36 hours. The tree can power MP3 players, cell phones and tablets through USB connection.

3D design for more efficient solar cells

Using a 3D design, sunlight can be trapped inside a solar cell for longer duration, allowing it more time to convert light into electricity. This breakthrough innovation comes from a company named Solar3D. Inspired by light management techniques used in fibre-optic devices, the innovating solar cell technology funnels light into silicon and keeps it trapped.

Silicon has a theoretical maximum light-to-electricity conversion efficiency of 29 per cent. The electrons generated